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Software Test Plan

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1. Introduction

1.1 Scope

This plan covers all verification and validation activity for the initial release of NOVA. It addresses two distinct concerns which are deliberately kept separate:

Concern	Question answered	Nature	Frequency
Software testing	Does the code behave as specified?	Deterministic, pass or fail	Every commit
Agent evaluation	Is the generated output of acceptable quality?	Statistical, threshold-based	On configuration change and on demand

Conflating the two produces an unstable pipeline and metrics that cannot be acted upon. Separation is what permits the regression gate to be trusted. A gate that fails for reasons the engineer cannot act upon is disabled in practice, and a disabled gate is worse than no gate because it presents as protection.

Out of scope: performance testing beyond the latency measurements defined in section 5.5; testing of the inference runtime itself; testing of any post-initial-release capability.

1.2 References

Reference	Title
NOVA-SRS-001	Software Requirements Specification
NOVA-SAD-001	Software Architecture Document

Reference	Title
NOVA-TM-001	Threat Model
NOVA-RR-001	Risk Register
ISO/IEC/IEEE 29119-3:2021	Software testing, test documentation

1.3 Glossary

See NOVA-SRS-001 section 1.4.

2. Context of the testing

2.1 Project

Single-engineer project, approximately 130 to 180 hours, delivered across seven milestones. Testing is not a milestone; it is concurrent with implementation and gated in continuous integration.

2.2 Test items

Item	Version under test
API service	Per commit
Worker and agent orchestration	Per commit
Memory subsystem: extraction, validation, storage, retrieval, lifecycle	Per commit
Strategy selection	Per commit
Web application	Per commit
Database schema and migrations	Per commit
Prompt and configuration versions	On change, via the evaluation gate
Deployment composition	Per release

2.3 Test scope

Included. All requirements of priority Must in NOVA-SRS-001. All quality scenarios QS-1 to QS-9 in NOVA-SAD-001. All threats T1 to T10 in NOVA-TM-001.

Excluded, with rationale.

Excluded	Rationale
Load and stress testing	Single user, single concurrent task. No meaningful load profile exists
Compatibility testing across browsers	One user, one browser. Stated as a limitation rather than tested
Localization testing	Single locale

Excluded	Rationale
Accessibility conformance audit	Not claimed. Basic semantic markup and screen-reader-associated form errors are required by NFR, but no conformance level is asserted
Inference runtime internals	Third-party dependency, addressed through contract tests at its boundary

2.4 Assumptions and constraints

ID	Assumption or constraint	Effect on testing
TA-1	A fixed random seed renders local inference sufficiently reproducible	If false, run-over-run comparison and the regression gate both become unreliable. Verified early
TA-2	Approximately 50 evaluation cases yield sufficient signal for gating	If false, the dataset is enlarged. Thresholds are not loosened
TA-3	Deterministic checks capture sufficient quality signal	If false, the share of model-based review increases and the reliability caveat is documented
TC-1	No external reviewer is available	Automated gates substitute for peer review where they can. The limitation is recorded, not concealed
TC-2	Evaluation suite duration must remain within the continuous integration budget	Constrains dataset size and the frequency of the expensive gate
TC-3	Evaluation data is synthetic and open-source only	The dataset is committable and reviewable

2.5 Stakeholders

Stakeholder	Interest in testing
Implementing engineer	Defect detection, regression prevention, confidence to refactor
Technical reviewer	Evidence that quality control is enforced rather than asserted

3. Testing communication

Single-engineer project. Defects are recorded as repository issues with a severity classification. Continuous integration results are the primary communication channel. Evaluation run results are persisted and trended over time.

The manual exploratory session findings in section 6.2 are recorded as issues rather than corrected silently, so that the record of systematic testing is itself an artifact.

4. Risk register

4.1 Product risks

ID	Risk	Likelihood	Impact	Mitigating test activity
PR-	Cross-tenant data disclosure	Low	Critical	Isolation suite as an unconditional gate

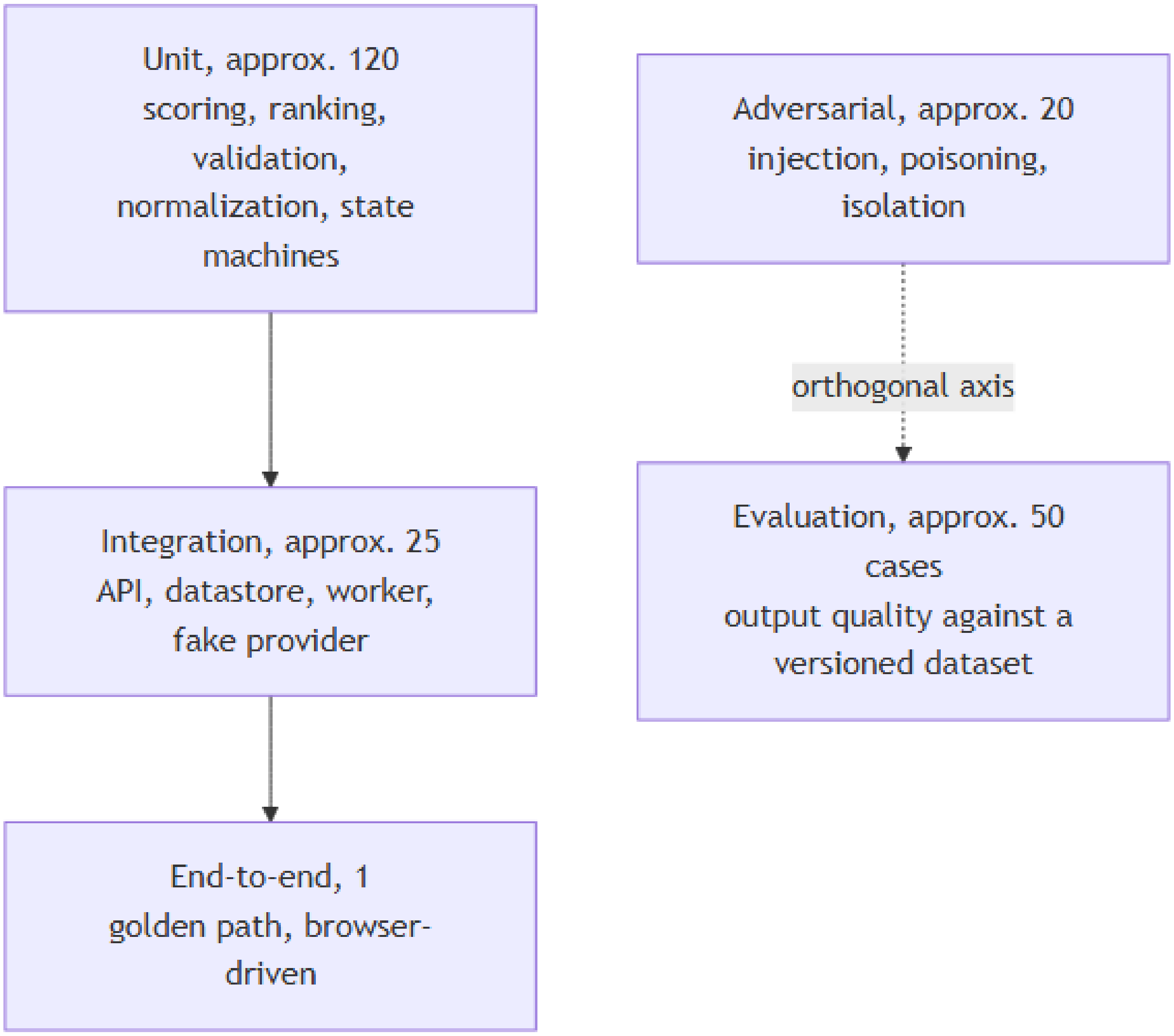
ID	Risk	Likelihood	Impact	Mitigating test activity
1				from the first milestone in which memory exists
PR-2	An injected instruction persists into stored memory	Medium	High	Adversarial scenarios targeting the extraction and confirmation path specifically, not only the generation prompt
PR-3	Adversarial feedback alters strategy selection	Low	Medium	Poisoning scenarios asserting score clamping and that no single event alters a selection
PR-4	A memory deletion does not take effect	Low	High	Deletion completeness test asserting both non-retrievability and absence from subsequent prompts
PR-5	A configuration change degrades output without detection	Medium	High	Regression gate demonstrated against a deliberately degraded prompt version
PR-6	Retrieval silently returns nothing and the system appears healthy	Medium	High	Explicit failure required on embedding unavailability. Asserted by test
PR-7	A failed task leaves no diagnostic record	Medium	Medium	Trace completeness test executed against a deliberately failed task

4.2 Project risks

ID	Risk	Mitigating activity
PJ-1	Evaluation dataset construction is underestimated	Authoring begins two milestones before the dataset is required. Twenty cases already exist from the model selection spike
PJ-2	Inference nondeterminism destabilizes the gate	Run-over-run variance measured before any threshold is set
PJ-3	Test effort is displaced by feature work under schedule pressure	Isolation tests and the regression gate are excluded from the pre-committed scope reduction order

5. Test strategy

5.1 Sub-processes



Sub-process	Objective	Entry criteria	Exit criteria
Unit	Verify logic in isolation	Component implemented	All pass. Statement coverage at least 80% on memory, retrieval, scoring, and validation logic
Integration	Verify component interaction and persistence	Unit tests passing, schema migrated	All pass against a real datastore and a fake inference provider
End-to-end	Verify the primary user journey through the	Integration passing	The golden path completes

Sub-process	Objective	Entry criteria	Exit criteria
	interface		
Adversarial	Verify security properties	Memory subsystem present	Zero failures. No tolerance
Evaluation	Quantify output quality and detect regression	Dataset versioned, baseline established	No gate metric below its threshold

5.2 Test design techniques

Technique	Applied to
Equivalence partitioning and boundary value analysis	Requirement length limits, score clamping bounds, exploration rate bounds, confidence floors
State transition testing	Task state machine, memory lifecycle, configuration version states, job lease and reclamation
Decision table testing	Feedback normalization across event types; retrieval inclusion given scope, tenant, pin, and threshold
Error guessing and exploratory testing	Manual charter sessions, section 6.2
Fault injection	Worker termination mid-task, inference provider unavailability, datastore unavailability, poison job
Security testing	Injection, privilege and isolation, feedback poisoning, output handling
Contract testing	Interface description drift; both inference provider implementations against a shared suite

Unit testing is where coverage is measured, because it is where the logic resides. The scoring functions written for the model selection spike already carry an assertion-based self-test, established before those functions were trusted with real output.

Integration testing uses a fake inference provider deliberately. These tests exist to verify wiring and must therefore be fast and deterministic. Model behaviour belongs to the evaluation sub-process. Combining the two yields a slow suite failing for reasons that cannot be acted upon.

End-to-end testing is confined to a single path. Such tests are expensive to maintain and the product is a single-user local application.

5.3 Test deliverables

Deliverable	Form
Automated test suites	Source, in the repository
Evaluation dataset	Versioned by content hash, in the repository

Deliverable	Form
Adversarial scenario definitions	Declarative configuration, in the repository
Evaluation run records	Persisted rows with per-metric aggregates and a gate verdict
Defect records	Repository issues with severity
Exploratory session notes	Repository issues
This plan	NOVA-STP-001

5.4 Test data requirements

Requirement	Detail
Composition	Approximately 50 cases, stratified across the four requirement types
Origin	Approximately 30 hand-authored, approximately 20 derived from open-source issue trackers and public specifications
Content per case	Requirement text, requirement type, expected properties (required case types, enumerated acceptance criteria, must-not-contain assertions), and labelled relevant memories for retrieval scoring
Held-out portion	A reserved subset never used for prompt tuning, so that the evaluation set does not become a tuning set
Versioning	Content hash. Any edit produces a new version. Comparison across versions is refused by the tooling
Sensitivity	Synthetic and open-source only. Committable and reviewable

Twenty cases already exist, produced for the model selection spike, five per requirement type, each with four machine-checkable acceptance criteria. They transfer directly and remove approximately one third of the authoring effort.

The hand-authored majority is deliberate. Derived cases contribute realism, but only hand-authored cases permit expected properties to be stated precisely enough for deterministic scoring. Where a metric must carry meaning, precision is worth more than realism.

5.5 Metrics to be collected

Each metric carries a definition, a collection method, a threshold, and a stated limitation. No measured value exists for most. Every threshold below is provisional.

ID	Metric	Definition	Threshold	Limitation
M1	Task success rate	completed Ã· (total âˆ’ user cancelled)	â‰¥95%	Carries no quality information. A completed poor plan counts as success. Not to be quoted as a quality figure
M2	Structured	valid on first attempt	â‰¥98%	Measures structure only. Observed 12

ID	Metric	Definition	Threshold	Limitation
	output validity	Ã· total generations. Also reported after repair	first attempt	of 12 over a 12-generation sample, which is far too small to establish a rate
M3	Acceptance criteria coverage	criteria with at least one linked case Ã· total criteria	â‰¥90%	Requires machine-parseable criteria. Relies on the model's own linking, so it is an upper bound
M4	Required case type presence	mean over plans of types present Ã· types required	â‰¥85%	Relies on the model's own case type labelling. A mislabelled case inflates the figure. Periodic manual audit required
M5	Duplicate case rate	cases exceeding within-plan cosine similarity Î¸ Ã· total cases	â‰¤10%	Î¸ is arbitrary and must be fixed and documented. Legitimately similar cases may be flagged
M6	Correction Recurrence Rate	confirmed corrections recurring as errors on later structurally similar tasks Ã· total confirmed corrections	â‰¤20%	Wholly dependent on the structural similarity threshold. Cannot distinguish a memory retrieved and disregarded from one never retrieved. Always reported alongside M7
M7	Retrieval precision and recall at 5	against the labelled relevance set	â‰¥70% / â‰¥60%	Labels represent one engineer's judgment over a small set. Relevance is partly subjective
M8	Evaluation pass rate	cases passing every gate Ã· total cases	â‰¥85%	Bounded by dataset breadth. A narrow dataset yields a flattering figure
M9	Provider call failure rate	failed calls Ã· total calls	â‰¤2%	Local failures are predominantly environmental, so this tracks environment health more than code health
M10	Latency percentiles	50th, 95th, 99th, end to end and per step	Set from data	Meaningless without stating GPU, model, and quantization. Cold model load reported separately: one measurement showed 115 s cold against 18 to 21 s warm for the same model
M11	Resource cost per task	tokens and elapsed time locally; billed cost for hosted providers	Not applicable locally	Local inference carries no marginal token cost. A monetary figure would be fabricated
M12	Regression rate	cases passing in run Nâˆ'1 and failing in run	â‰¤5%	Comparable only within a dataset version. Sensitive to nondeterminism,

ID	Metric	Definition	Threshold	Limitation
		N Ã· cases passing in run Nâˆ’1		mitigated by a fixed seed, with residual variance measured before thresholds are set

M6 is the primary metric. It renders the project's central claim falsifiable and is reported including any run in which it deteriorates.

5.6 Test completion criteria

Sub-process	Completion criterion
Unit	All pass. Statement coverage at least 80% on memory, retrieval, scoring, and validation logic
Integration	All pass, including worker termination, lease reclamation, poison job, and failed-task trace persistence
End-to-end	The golden path completes through the interface
Adversarial	Zero failures. Cross-tenant isolation admits no tolerance
Evaluation	No gate metric below threshold. M6 measured across at least two runs on a fixed dataset version

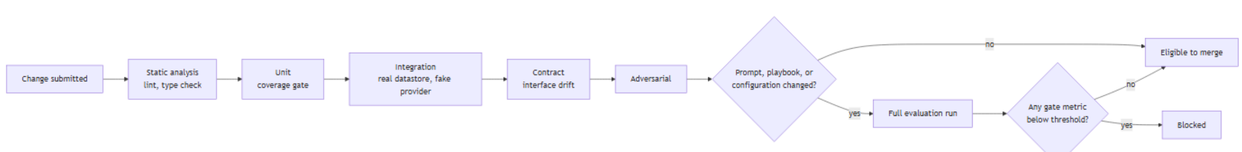
Release acceptance additionally requires: a deliberately degraded prompt version demonstrably blocked by the gate; deployment from a clean checkout on a separate host; and a completed restore drill.

5.7 Test environment requirements

Environment	Composition
Development	Local. Real datastore in a container, fake inference provider
Continuous integration	Ephemeral datastore container. Fake provider for integration; pinned real model for evaluation
Evaluation	Pinned model version, fixed seed, **separate datastore instance**. The evaluation harness must never share a datastore with a working instance

The reference configuration for every timing measurement is: NVIDIA RTX 4050 Laptop, 6 GB VRAM, 4B-parameter model at 4-bit quantization, 8192-token context, inference runtime on the host. Measurements are void without this qualification.

5.8 Retesting and regression testing



Gate	Blocks merge
Static analysis, unit, integration, contract	Yes
Adversarial suite, any failure	Yes, unconditionally
Cross-tenant isolation	Yes, unconditionally
Full evaluation run	Only where prompts, playbooks, or configuration changed
Any gate metric below threshold	Yes

The expensive gate executes only where it can produce information. Thresholds derive from measured variance rather than from intent; that measurement does not yet exist and is a prerequisite recorded in NOVA-SDP-001.

Every defect of severity S1 or S2 receives a failing test before the corrective change. For defects of output quality, the regression test is a new evaluation case added to the dataset, which is how the dataset accrues coverage rather than being authored once and left static.

5.9 Suspension and resumption criteria

Condition	Action
Cross-tenant disclosure observed	Suspend feature work. Correct, add a regression test, resume
Regression gate observed to flap	Suspend reliance on the gate. Measure variance, reset thresholds, resume
Evaluation dataset edited without a version increment	Suspend trend reporting until the version is corrected
Inference environment altered	Suspend cross-run comparison until a new baseline is established

6. Testing activities and estimates

6.1 Automated activity

Activity	Milestone	Estimate
Unit suites, developed alongside implementation	M2 to M5	Included in feature estimates
Integration suite	M2 to M3	6 h
Isolation suite	M3	3 h
End-to-end path	M3	2 h
Evaluation dataset authoring	M2 to M5	15 to 25 h
Evaluation harness and metric computation	M5	10 h
Adversarial scenario suite	M5	6 h
Continuous integration configuration and gating	M2, M5	4 h

6.2 Manual activity

Charter-based, time-boxed to approximately 60 minutes each. Automation does not detect everything.

Charter	Objective
Authentic usage	Operate NOVA on genuine requirements from personal work. Record every point of friction
Memory management	Deliberately construct contradictory, overlapping, and nonsensical memories. Observe system behaviour
Trace comprehension	Present the trace viewer to a reader unfamiliar with the system. Determine whether they can explain what occurred. If they cannot, the observability provision is decorative
Manual adversarial	Attempt injections the automated suite does not cover
Cold start	Fresh database. Assess whether a first-time experience is coherent
Failure presentation	Terminate the inference runtime mid-task. Assess whether the resulting error is actionable

6.3 Production-like validation

No production environment exists. The nearest equivalents, described as such:

- Deployment from a clean checkout on a separate host or virtual machine, which detects the class of defect that appears only in the absence of accumulated development-machine state
- Sustained run of at least 50 sequential tasks, monitoring memory growth, trace volume, and retrieval degradation
- Restore drill from backup
- Reproducibility check: identical seed, model version, and configuration version yielding equivalent output. Should this fail, every regression comparison is noise

7. Staffing

Role	Held by	Responsibilities
Test manager	Laxmi Poudel	This plan, gate definitions, threshold setting
Test designer and implementer	Laxmi Poudel	Suite construction, dataset authoring, scenario definition
Test executor	Continuous integration, plus the engineer for manual charters	Execution, result recording

No hiring or training need arises. The single-reviewer limitation is recorded as project risk PJ-3 and as constraint TC-1.

8. Schedule

Milestone	Testing activity	Gate introduced
M1	Spike harness self-test	None
M2	Unit and integration suites, contract test, initial dataset cases	Static analysis, unit, integration, contract
M3	Isolation suite, deletion completeness, end-to-end path, extraction precision measurement	Isolation, unconditional
M4	Selection determinism and poisoning scenarios	Included in the adversarial suite
M5	Full dataset, evaluation harness, adversarial suite, variance measurement, threshold setting	Adversarial and evaluation regression gate
M6	Fault injection for every degraded mode, soak run, restore drill, clean-checkout deployment	Log content assertion
M7	Documentation accuracy pass, tracing every stated claim to a test, metric, or decision record	None

9. Defect classification

Severity	Definition	Response
S1	Data loss, cross-tenant disclosure, unbounded resource consumption	Suspend feature work. Correct, add a regression test, resume
S2	Primary workflow inoperable	Correct within the current milestone
S3	Degraded quality or confusing presentation	Backlog with a recorded issue
S4	Cosmetic	Backlog. May remain uncorrected

Revision history

Version	Date	Author	Change
0.1	2026-09-08	Laxmi Poudel	Initial draft